



CELEBAL

TECHNOLOGIES

**Azure Cloud Fundamentals and Data Pipeline
Implementation using ADF**

G Rithwik Surya

Task 1:

1. Explore Azure Portal :

- Logged into microsoft azure using a student account

2. Create a Resource Group :

A resource group is created in azure to organize and manage all resources required for the project. Resource groups helps in management and monitoring

- Created a resource group with ane 'week4assignmentmtdatfactory'
- Selected the student plan subscription and 'east asia'

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, a search bar, and user account information for '23B81A05AT@cvr.ac.in'. The main content area is titled 'Resource Manager | Resource groups' and shows a list of resource groups. A single resource group is visible:

Name	Subscription	Location
week4assignmentmtdatfactory	Azure for Students	East Asia

The interface also includes a left-hand navigation pane with options like 'Resource Manager', 'All resources', 'Favorite resources', 'Recent resources', 'Resource groups', 'Tags', 'Organization', 'Tools', 'Deployments', and 'Help'. At the bottom, it indicates 'Showing 1 - 1 of 1. Display count: auto' and provides a 'Give feedback' link.

Task 2: Storage Setup

1. Create a Storage Account :

A storage account is a cloud storage service in azure used to store and manage data

- Created a new storage account with name as 'week4assignmentstorage'
- Assigned this storage account to the resource group that i had already created 'week4assignmentmtdatafactory'
- Selected the student plan and east asia region
- Selected azure blob storage or azure data lake storage
 - Blob storage is used because as we are storing csv files blob storage is the best and the effective option as it allows to store unstructured data and also its easy to integrate with azure data factory

Home > Storage center

Storage center | Blob Storage

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Search

Summary Resources

Overview

All storage resources

Object storage

Blob Storage

File storage

Block storage

Azure Disks

Elastic SAN

Data management

Migration

Partner solutions

Management services

Help

Create Restore Manage view Refresh Export to CSV Open query Assign tags Delete Add to service group Group by none

Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter

Name	Type	Kind	Resource Group	Location	Subscription
week4assignmentstorage	Storage account	StorageV2	week4assignmentdatafa...	East Asia	Azure for Students

Showing 1 - 1 of 1. Display count: auto

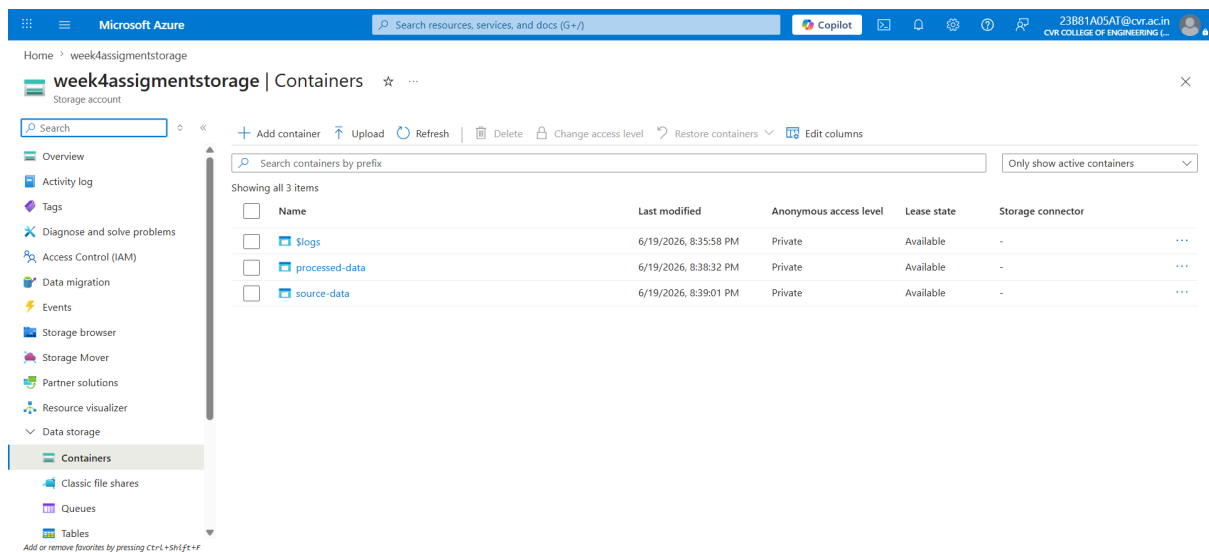
Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

2. Create a Blob Container :

A blob container is within azure blob storage used to store the files and organize them ,multiple files could be uploaded and managed in a container.

- Created two containers as source-data and processed-data one to store the raw files and another one is to store the copied file



3. Upload a CSV file :

We need to upload the dataset into the containers

- Uploaded the sample_superstore.csv dataset into the source-data
- And created a sample file named as cleaned_dataset inside processed-data to store the copied file
- Along with these two files also uploaded 3 files named as customer1 ,customer 2 and customer 3 to run for each activity in adf

Source-data

Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > week4assignmentstorage | Containers

source-data

Container

Search

+ Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

source-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 4 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - Superstore.csv	6/19/2026, 8:39:35 PM	Hot (Inferred)	Block blob	2.18 MiB	Available
☐	customer1.csv	6/19/2026, 9:30:02 PM	Hot (Inferred)	Block blob	157 B	Available
☐	customer2.csv	6/19/2026, 9:30:09 PM	Hot (Inferred)	Block blob	148 B	Available
☐	customer3.csv	6/19/2026, 9:30:17 PM	Hot (Inferred)	Block blob	161 B	Available

Add or remove favorites by pressing Ctrl+Shift+F

Processed-data

Microsoft Azure

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Home > week4assignmentstorage | Containers

processed-data

Container

Search

+ Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

processed-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	cleaned_dataset.csv	6/20/2026, 10:00:32 AM	Hot (Inferred)	Block blob	148 B	Available

Add or remove favorites by pressing Ctrl+Shift+F

Task 3: ADF Basics

1. Create Azure Data Factory :

Azure data factory is a cloud based data integration service that enables users to create and manage pipelines. It is a cloud based elt and etl service

- Created a azure data factory with the name 'adfweek4assignment'
- Selected the students account and resource group as 'week4assignmentdatafactory ' where the region was east asia same as resource group

Microsoft Azure

Search resources, services, and docs (G+/)

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Home

Data factories ☆ ...

Identify non-compliant Data factories in my environment. Group Data factories by location in ARG query. Sort Data factories by creation date.

+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group Group by none

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

☐	Name ↑	Type	Subscription	Resource Group	Location	
☐	adfweek4assignment	...	Data factory (V2)	Azure for Students	week4assignmentdatafactory	East Asia

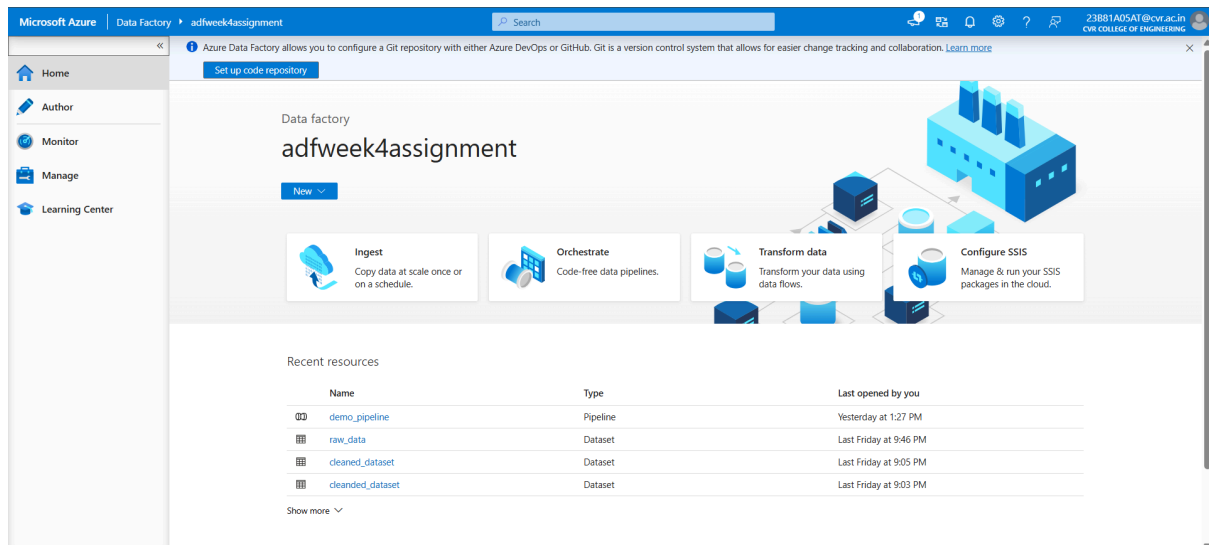
Showing 1 - 1 of 1. Display count: auto

Give feedback

2. Explore ADF UI (Author, Monitor, Manage) :

- **Author:** Used to create and design data pipelines, datasets, linked services, and data flows in Azure Data Factory
- **Monitor:** Used to track pipeline executions and identify errors during execution.

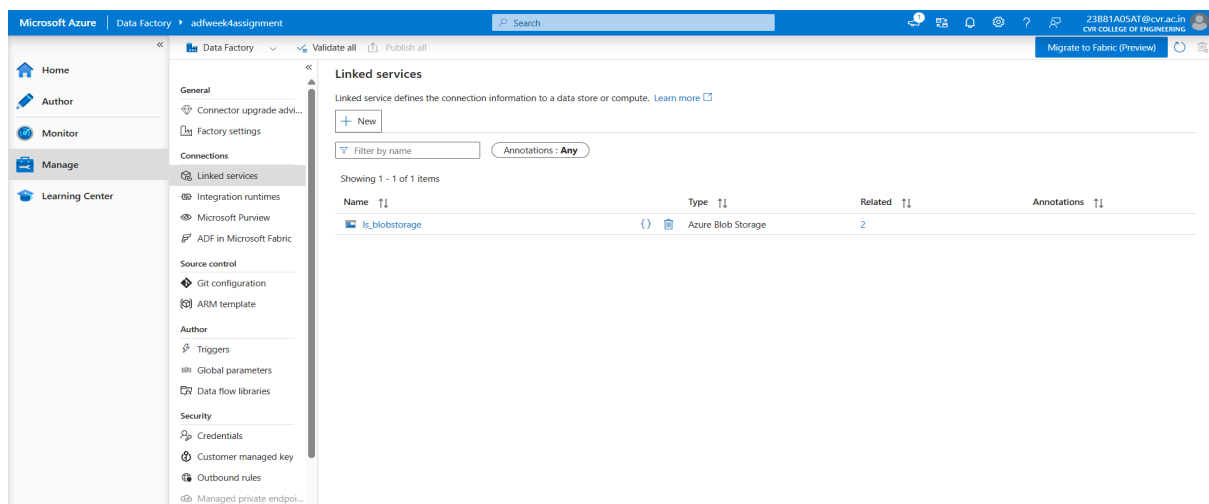
- **Manage:** Used to configure connections, integration runtimes triggers and other settings in Azure Data Factory.



3.Create Linked Service (Blob Storage) :

Linked Service is a connection configuration in Azure Data Factory that allows ADF to connect to external data sources and destinations

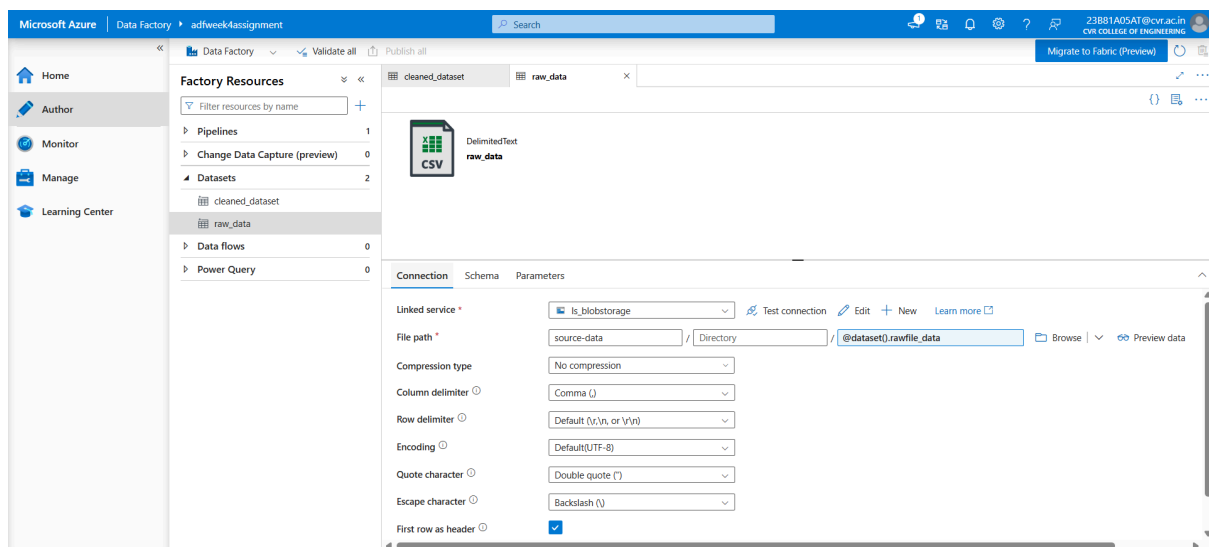
- For creating a linked service selected azure blob storage datastore and names the service as 'ls_blobstorage'
- Selected subscription for students
- To test whether the linked service working or not click on test connection and apply

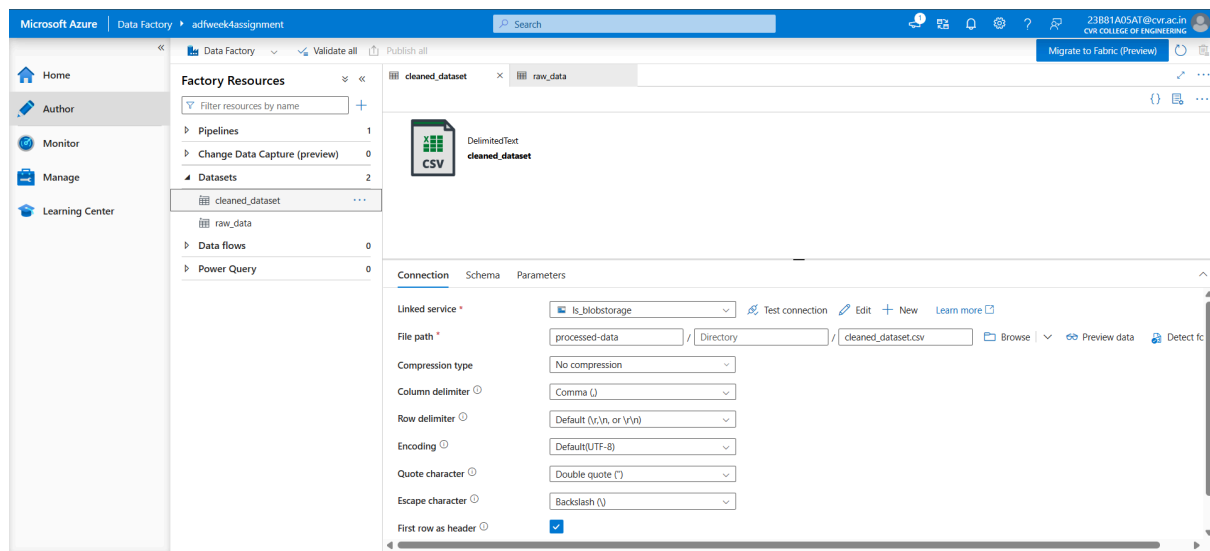
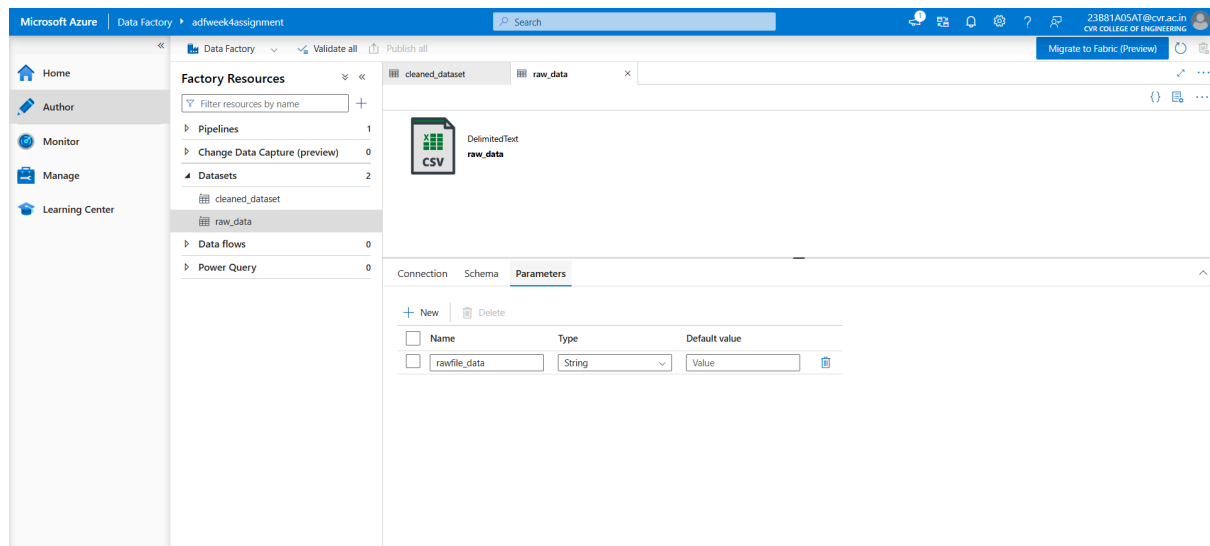


4. Create datasets (source + destination) :

To create datasets we need to go to author and click on datasets

- Selected azure blob storage for data store type and as the data i uploaded is a csv file selected delimitedText for the file type
- And gave the names as raw_data for one and another one is cleanded_dataset and added the ls_blobstorage link service to them.
- Added the parameters to configure multiple files
- Configured the file path for both datasets. For the raw_dataset, since multiple files need to be processed dynamically, the file name was parameterized using `@dataset().rawfile_data`



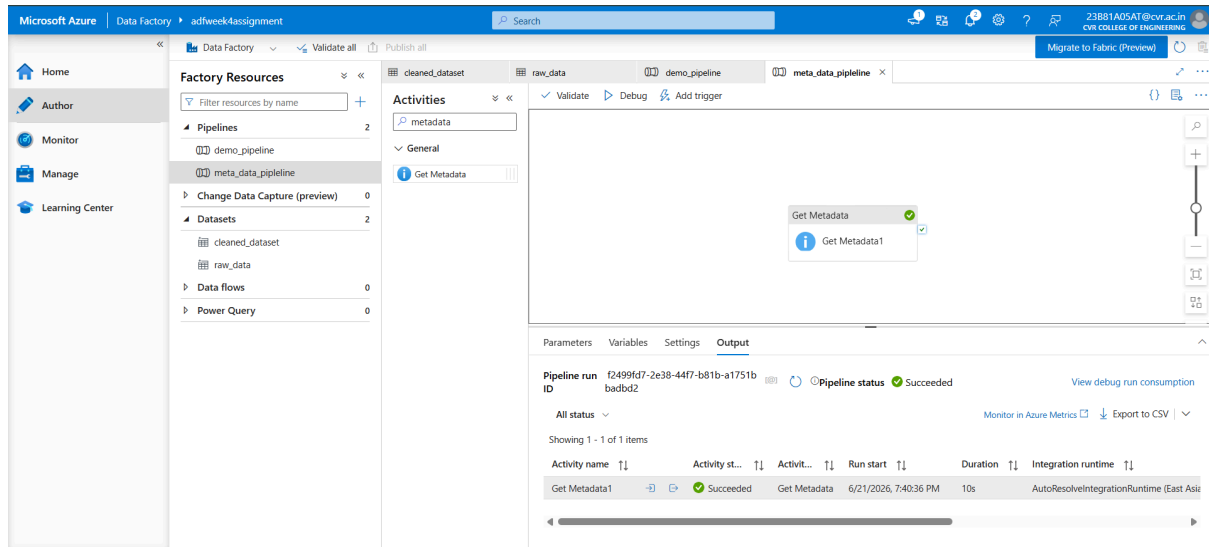


5. Use Get Metadata activity :

Get Metadata Activity is used to retrieve information about a file or folder before processing it in a pipeline

- Created a pipeline and named it as meta_data_pipeline
- Added multiple fields as
- Exists, last modified and item name

- And published everything after that to check clicked on debug
- It showed as succeeded



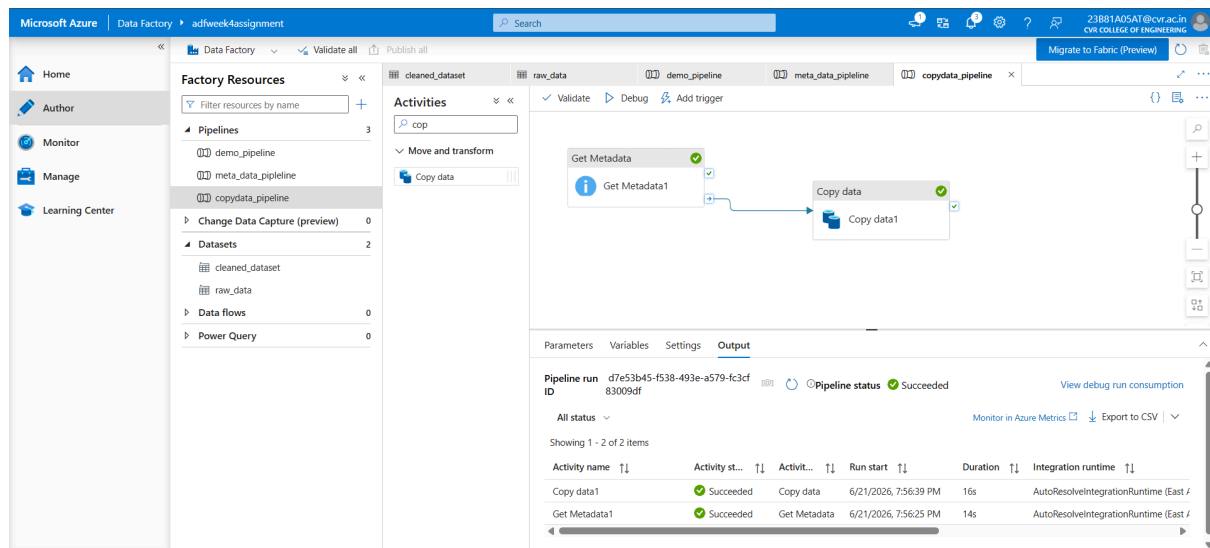
Task 4: Pipeline Development:

1. Create pipeline using Copy Data activity :

Copy Data Activity is used to transfer data from a source dataset to a destination dataset within Azure Data Factory. It enables automated data movement between different storage systems and services

2. Configure source and destination :

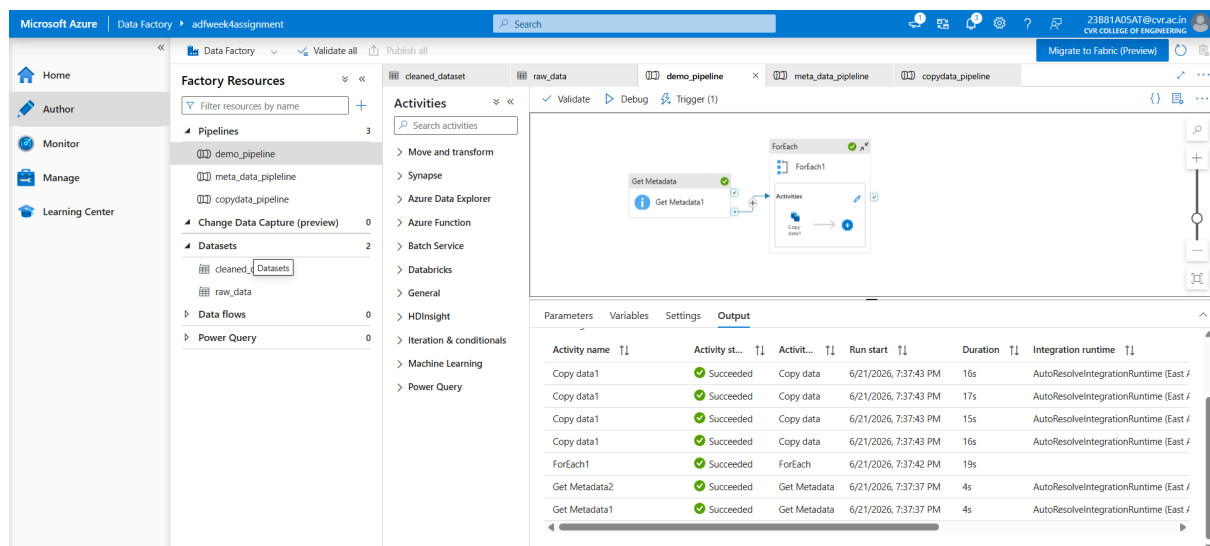
for copy activity we need to configure source and sink with source file and destination file where in my case it is source-data and processed-data



3. Add ForEach activity (optional if multiple files) :

ForEach Activity is used to iterate through a collection of items and execute one or more activities repeatedly for each item.

- Select for each activity and configured the parameter to run the multiple files where i selected meta data foreach and copy activity to run this pipe line



Task 5: Pipeline Execution

1.Run pipeline using Debug/Trigger

To run the pipeline there are two ways we can add a trigger there are three types of trigger used according to the need and we can also run using debug
If the pipeline run without any errors it displays as succeeded with outputs

Outputs of 3 pipelines when ran:

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists Pipelines (demo_pipeline, meta_data_pipeline, copydata_pipeline) and Datasets (cleaned_dataset, raw_data). The 'Activities' pane on the right lists various services like Move and transform, Synapse, Azure Data Explorer, etc. The main canvas displays a single activity named 'Get Metadata1' with a green checkmark indicating success. Below the canvas, the 'Output' tab shows the pipeline run details: Pipeline run ID d865d90-9d31-4251-81ae-6a155eb0c0d4, Pipeline status Succeeded, and a table with one row for the 'Get Metadata1' activity, which is marked as 'Succeeded'.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Get Metadata1	✓ Succeeded	Get Metadata	6/21/2026, 9:30:53 PM	13s	AutoResolveIntegrationRuntime (East Asi

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists Pipelines (demo_pipeline, meta_data_pipeline, copydata_pipeline) and Datasets (cleaned_dataset, raw_data). The 'Activities' pane on the right lists various services like Move and transform, Synapse, Azure Data Explorer, etc. The main canvas displays a sequence of two activities: 'Get Metadata1' and 'Copy data1', both with green checkmarks indicating success. Below the canvas, the 'Output' tab shows the pipeline run details: Pipeline run ID d7e53b45-f538-493e-a579-fc3cf83009df, Pipeline status Succeeded, and a table with two rows for the activities, both marked as 'Succeeded'.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	6/21/2026, 7:56:39 PM	16s	AutoResolveIntegrationRuntime (East /
Get Metadata1	✓ Succeeded	Get Metadata	6/21/2026, 7:56:25 PM	14s	AutoResolveIntegrationRuntime (East /

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Factory Resources

- Pipelines
 - demo_pipeline
 - meta_data_pipeline
 - copydata_pipeline
- Datasets
 - cleaned_dataset
 - raw_data
- Data flows
 - 0
- Power Query
 - 0

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

demo_pipeline

Get Metadata1

ForEach1

Parameters Variables Settings Output

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	6/21/2026, 7:37:43 PM	16s	AutoResolveIntegrationRuntime (East /
Copy data1	✓ Succeeded	Copy data	6/21/2026, 7:37:43 PM	17s	AutoResolveIntegrationRuntime (East /
Copy data1	✓ Succeeded	Copy data	6/21/2026, 7:37:43 PM	15s	AutoResolveIntegrationRuntime (East /
Copy data1	✓ Succeeded	Copy data	6/21/2026, 7:37:43 PM	16s	AutoResolveIntegrationRuntime (East /
ForEach1	✓ Succeeded	ForEach	6/21/2026, 7:37:42 PM	19s	AutoResolveIntegrationRuntime (East /
Get Metadata2	✓ Succeeded	Get Metadata	6/21/2026, 7:37:37 PM	4s	AutoResolveIntegrationRuntime (East /
Get Metadata1	✓ Succeeded	Get Metadata	6/21/2026, 7:37:37 PM	4s	AutoResolveIntegrationRuntime (East /

Scheduled trigger:

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Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Chennai, Kolkata, Mu... : Last 24 hours Pipeline name: All Status: All Runs: Latest runs Triggered by: All

Add filter

Showing 1 - 1 items

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters	Annotations	Run ID
demo_pipeline	6/21/2026, 10:00:00 AM	6/21/2026, 10:00:34 AM	34s	demo_schedule_trig...	✓ Succeeded	Original			78

Last refreshed 0 minutes ago

The pipeline ran at 21/6 10:00 am as scheduled before

Debug pipeline runs

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Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Chennai, Kolkata, Mu... : Last 24 hours Pipeline name: All Status: All Add filter

Showing 1 - 7 items

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
meta_data_pipeline	6/21/2026, 9:30:44 PM	6/21/2026, 9:31:07 PM	23s	✓ Succeeded	Manual trigger	d865d90-9d31-4251-81ae...	
copydata_pipeline	6/21/2026, 7:56:18 PM	6/21/2026, 7:56:55 PM	38s	✓ Succeeded	Manual trigger	d7e53b45-f538-493e-a579...	
copydata_pipeline	6/21/2026, 7:54:01 PM	6/21/2026, 7:54:40 PM	40s	✗ Failed	Manual trigger	068e3c4-411c-42b2-97e2...	
copydata_pipeline	6/21/2026, 7:49:57 PM	6/21/2026, 7:50:20 PM	24s	✗ Failed	Manual trigger	2cf2271d-5418-4668-9dad...	
meta_data_pipeline	6/21/2026, 7:40:29 PM	6/21/2026, 7:40:47 PM	19s	✓ Succeeded	Manual trigger	f2499f47-2e38-4447-b81b...	
demo_pipeline	6/21/2026, 7:37:29 PM	6/21/2026, 7:38:01 PM	32s	✓ Succeeded	Manual trigger	d0a6f62b-ba5a-46f5-87a5...	
demo_pipeline	6/21/2026, 7:35:52 PM	6/21/2026, 7:36:33 PM	42s	✓ Succeeded	Manual trigger	7c01aaa1-aea7-4172-8892...	

Task 6: IAM Roles

1. Assign roles:

- Reader
- Contributor

IAM (Identity and Access Management) is used in Azure to control who can access resources and what actions they can perform. It helps secure Azure resources by assigning appropriate permissions

The Reader role provides read-only access to Azure resources.

The Contributor role allows users to manage Azure resources.

- To assign the role go to the iam section and click on add role assignment and assign the roles ‘
- But here i have an issue that i cannot assign contributor role due to student account

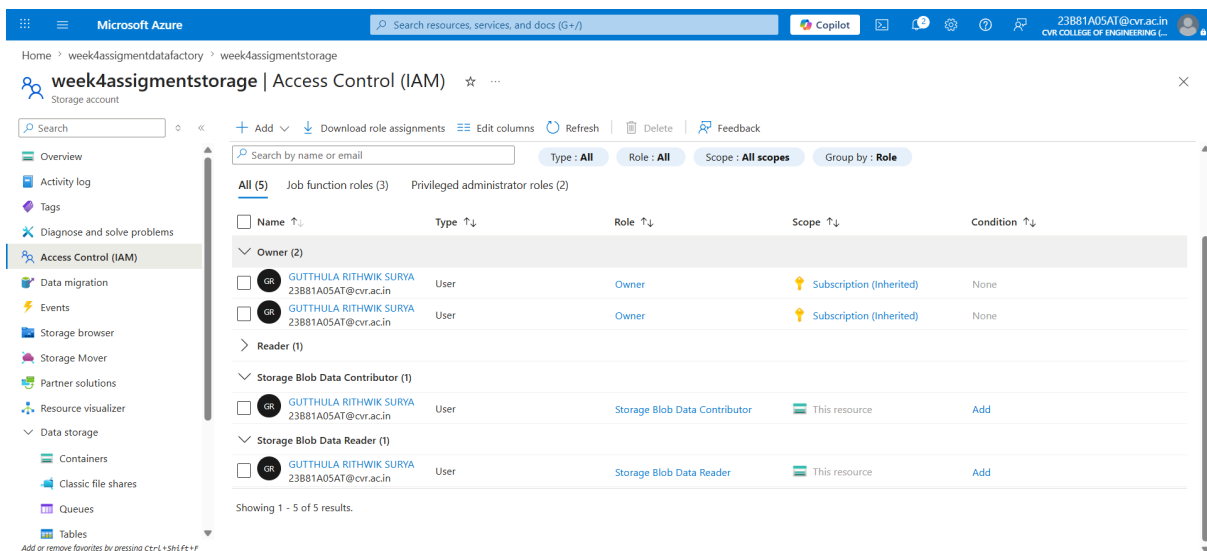
The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is titled "week4assignmentdatafactory | Access control (IAM)". The left sidebar shows the navigation menu with "Access control (IAM)" selected. The main area displays the "Number of role assignments for this subscription" as 6 out of 4000, and "Privileged" as 2. Below this, there is a search bar and filters for "Type: All", "Role: All", "Scope: All scopes", and "Group by: Role". The table lists role assignments:

Name	Type	Role	Scope	Condition
Owner (2)				
GUTTHULA RITHWIK SURYA 23B81A05AT@cvr.ac.in	User	Owner	Subscription (Inherited)	None
GUTTHULA RITHWIK SURYA 23B81A05AT@cvr.ac.in	User	Owner	Subscription (Inherited)	None
Reader (1)				
GUTTHULA RITHWIK SURYA 23B81A05AT@cvr.ac.in	User	Reader	This resource	None

Showing 1 - 3 of 3 results.

Provide access to ADF for Storage

To provide access to adf for storage we need to assign roles at data storage we can provide two roles as storage blob data contributor and storage blob data reader



During this assignment, I learned how different Azure services work together to build a complete data pipeline. I created a Resource Group, Storage Account, Blob Container, and Azure Data Factory instance, which helped me understand how resources are organized and managed in Azure. I also learned how to create Linked Services and Datasets to establish connections between Azure Data Factory and Azure Blob Storage.

While building the pipeline, I gained practical experience with Get Metadata, ForEach, and Copy Data activities. The most challenging part for me was configuring the ForEach activity. Initially, I was unable to process multiple files because I did not understand how dataset parameters worked and how values needed to be passed dynamically using expressions such as `@item().name`. I spent considerable time troubleshooting errors and understanding the relationship between Get Metadata output, ForEach iterations, and dataset parameters. Through this process, I learned how dynamic

content is used in Azure Data Factory and how pipelines can be designed to process multiple files automatically.

I also explored Azure IAM and understood the purpose of Reader and Contributor roles in controlling access to resources. Another important lesson was learning how to validate pipelines, monitor executions, identify errors from activity outputs, and debug failures effectively.

Overall, this assignment gave me hands-on experience with Azure cloud services and helped me understand the complete workflow of building, troubleshooting, and executing data pipelines. More than creating the pipeline itself, I learned how to solve configuration issues, work with dynamic parameters, and use Azure Data Factory to automate data movement in a real-world scenario.